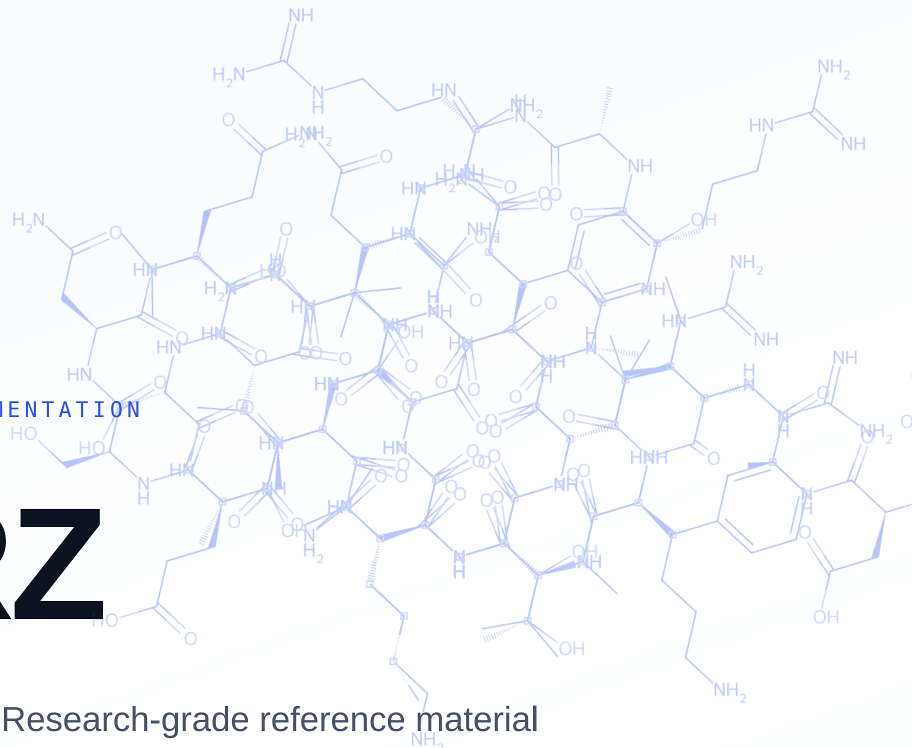
ANVIL COMPOUNDS RESEARCH USE ONLY

RESEARCH GRADE · 99%+ PURITY · USA-BASED



RESEARCH REFERENCE & SAFETY DOCUMENTATION

GLP-TRZ

Dual GIP / GLP-1 receptor agonist · Research-grade reference material

COMPOUND	CLASS	LENGTH	MW
Tirzepatide	**Dual incretin agonist**	**39 amino acids**	**≈4814 g/mol**

Identity & Composition

Tirzepatide is a synthetic 39-amino-acid peptide based on the native GIP sequence, modified with a C20 fatty diacid moiety that enables albumin binding and supports once-weekly pharmacokinetics. It is a single molecule that activates two receptors — GIP and GLP-1 — distinguishing it from single-receptor GLP-1 agonists.

Product name	GLP-TRZ (tirzepatide)
Class	Dual GIP / GLP-1 receptor agonist
Backbone	GIP-based synthetic peptide
Length	39 amino acids
Modification	C20 fatty diacid conjugate (albumin binding)
Molecular weight	≈4814 g/mol
Appearance	White lyophilized powder
Solubility	Soluble in water
Identified use	In-vitro research & development only

CRITICAL DISTINCTION

The tirzepatide molecule is an FDA-approved prescription medicine (Mounjaro, Zepbound). Research-grade material supplied as a reference standard is **not** that approved drug product.

It is not manufactured to pharmaceutical standards, carries no approved labeling, and must not be represented, dispensed, or used as a medicine. Supplied solely for in-vitro laboratory research.

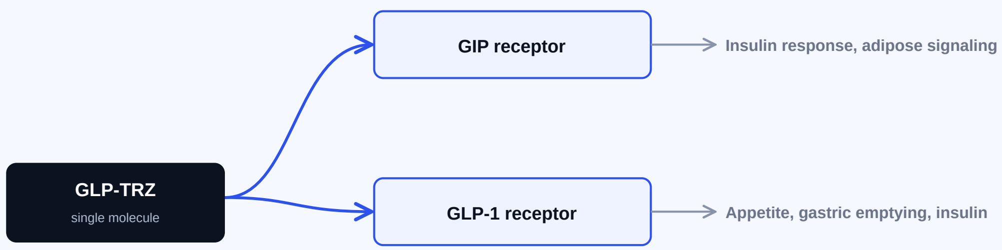
Note on class safety labeling. The approved product carries a boxed warning regarding thyroid C-cell tumours observed in rodents, and class-level considerations including pancreatitis and gallbladder events. These pertain to clinical use of the approved medicine and are noted here for scientific completeness.

Mechanism

GIP and GLP-1 are endogenous incretin hormones. Their receptors are expressed in pancreatic tissue and in brain regions relevant to appetite regulation. Tirzepatide co-activates both with a single molecule — the mechanistic feature that separates it from the GLP-1 monotherapy class.

Pharmacology

- **Dual receptor co-agonism.** Simultaneous activation of GIP and GLP-1 receptors, which have overlapping and non-overlapping expression and function.
- **Appetite and intake.** Reduced calorie intake, with effects understood to be substantially appetite-mediated.
- **Gastric emptying.** Slowed gastric emptying, a class effect of GLP-1 receptor agonism.
- **Insulin sensitivity and glycemic control.** Improved insulin response; the basis of the original type 2 diabetes indication.
- **Half-life extension.** The C20 fatty diacid moiety supports albumin binding and once-weekly administration in the approved product.



Dual incretin receptor co-agonism. Schematic. Research-grade material supplied here is for in-vitro laboratory use only.

What the Literature Supports — and What It Doesn’t

Unlike most compounds in this catalogue, tirzepatide is supported by a large, completed, peer-reviewed Phase 3 programme and holds multiple FDA approvals. The evidence is strong — and it pertains to the **approved medicine under clinical supervision**, not to research-grade material.

Evidence at a glance

PHASE 3 PROGRAMME	FDA STATUS	HEAD-TO-HEAD DATA	BOXED WARNING
SURPASS + SURMOUNT	Approved (3 indications)	Available	Thyroid C-cell (rodent)

- **SURMOUNT-1** (N=2,539, non-diabetic adults with obesity/overweight): mean weight reduction 15.0% / 19.5% / 20.9% at 5 / 10 / 15 mg versus 3.1% with placebo at 72 weeks.
- **SURMOUNT-5** head-to-head versus semaglutide: 20.2% versus 13.7% mean weight loss at 72 weeks; published in NEJM.
- **Approvals:** Mounjaro (May 2022, type 2 diabetes); Zepbound (Nov 2023, chronic weight management); Zepbound for moderate-to-severe obstructive sleep apnea with obesity (2024).
- **Tolerability:** predominantly gastrointestinal. In SURMOUNT-1, nausea 31.0% (15 mg) versus 11.7% placebo; discontinuation for adverse events 6.3%.
- **Ongoing research** in chronic kidney disease, heart failure with preserved ejection fraction, and morbidity/mortality in obesity.

> **Appropriate characterization:** a rigorously evidenced, FDA-approved dual incretin agonist. Any research use of reference-grade material is distinct from clinical use of the approved product, and clinical trial outcomes do not transfer to unapproved material of unverified provenance.

Regulatory & status notes

US FDA	Approved: Mounjaro (2022, T2D); Zepbound (2023, chronic weight management; 2024, OSA with obesity). Prescription medicine.
Safety labeling	Boxed warning for thyroid C-cell tumours (rodent findings); class considerations include pancreatitis and gallbladder events.
Supply basis	Research reagent for in-vitro laboratory use only; not the approved drug product; not for human or veterinary administration.

Selected references

1. Jastreboff AM, et al. Tirzepatide once weekly for the treatment of obesity (SURMOUNT-1). N Engl J Med. 2022.
2. SURMOUNT-5: tirzepatide versus semaglutide, head-to-head Phase 3b (NCT05822830). N Engl J Med. 2025.
3. SURPASS Phase 3 programme in type 2 diabetes (five global registration trials).
4. Mounjaro and Zepbound US Prescribing Information. FDA/DailyMed.

Safety Data Sheet

Prepared per GHS Rev. 8 / OSHA HazCom 2012
(29 CFR 1910.1200) · 16-section format

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	GLP-TRZ (tirzepatide)
Class	Dual GIP / GLP-1 receptor agonist, 39 aa
MW	≈4814 g/mol
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use. Research-grade material is not an approved drug product.
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data for the research-grade substance; occupational toxicological data are limited. **Signal word:** None. **Pictograms:** None required. **Precautionary:** P261, P264, P280, P501. **Handle as a potent pharmacologically active peptide** — active at microgram-to-milligram quantities in vivo; minimize all exposure.

3 Composition

Single-substance product, >98% research grade. Residual synthesis reagents, solvents and counter-ions may be present consistent with research-grade material. See lot COA for impurity profile.

4 First-Aid Measures

Eyes: rinse with water. **Skin:** wash with soap and water; remove contaminated clothing. **Inhalation:** fresh air; seek medical advice if symptoms develop. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Inform medical personnel that the substance is an incretin receptor agonist; monitor for hypoglycaemia and gastrointestinal effects.

5 Fire-Fighting

Media: CO_2, dry chemical, foam, water spray. Combustion may produce CO, CO_2, nitrogen oxides and sulfur oxides. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust formation; ventilate; use PPE per §8. Prevent entry to drains/waterways. Collect in a closed labeled container for disposal per §13. Decontaminate the affected surface.

7 Handling & Storage

Handle as a potent pharmacologically active peptide; containment is important. Weigh and reconstitute in a fume hood or ventilated balance enclosure. Store lyophilized solid tightly closed, protected from light/moisture/heat; −20 °C long-term, 2–8 °C short-term. Aliquot reconstituted solutions; store frozen, protected from light; avoid repeated freeze-thaw. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Because the substance is pharmacologically potent, engineering containment when handling dry powder is a priority. Fume hood or ventilated balance enclosure; nitrile gloves (double-glove for neat powder); ANSI Z87.1 eye protection; lab coat; N100/P100 respirator per 29 CFR 1910.134 where controls are insufficient.

9 Physical & Chemical Properties

State	Lyophilized powder
Appearance	White powder
Odor	Odorless
Solubility	Soluble in water
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable as a lyophilized solid under recommended storage. Avoid heat, moisture, sunlight, UV, open flame. Incompatible with strong oxidizers, strong acids/bases, strong reducing agents. Decomposition: CO, CO_2, NOx, SOx. No hazardous polymerization.

11 Toxicological Information

Clinical safety information exists for the approved drug product under medical supervision and is documented in its labeling; **that information does not characterize occupational exposure** to research-grade substance. No occupational LD_{50}/LC_{50} identified. Skin/eye: treat as potentially irritating. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity, persistence or bioaccumulation data identified. Avoid release to surface water, soil, drains or sewers.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

The tirzepatide *molecule* is an FDA-approved prescription drug substance (Mounjaro, Zepbound). **Material supplied here is research grade: not an approved drug product, not manufactured to pharmaceutical standards, and must not be represented, dispensed or used as a medicine.** US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use. Sources: PubChem; FDA labeling; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8. Provided without warranty. For scientific R&D use only.